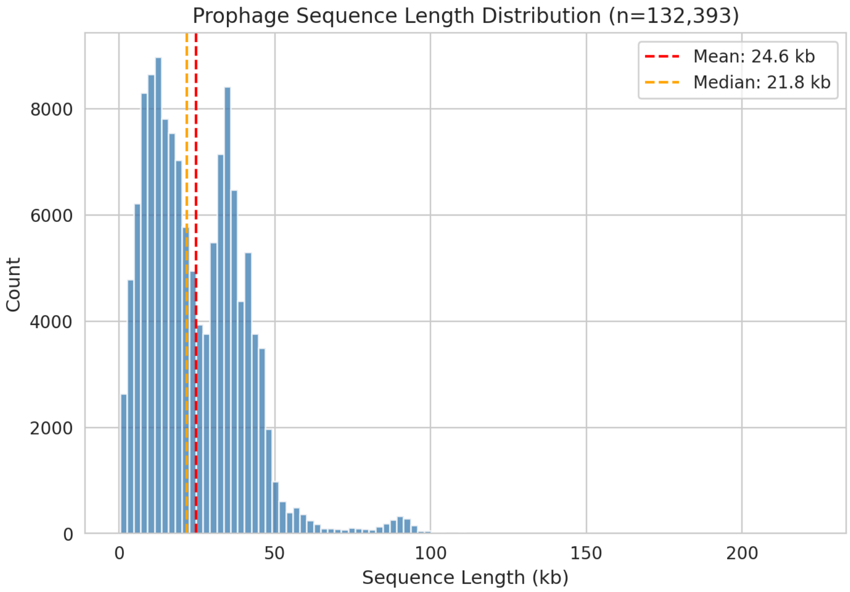
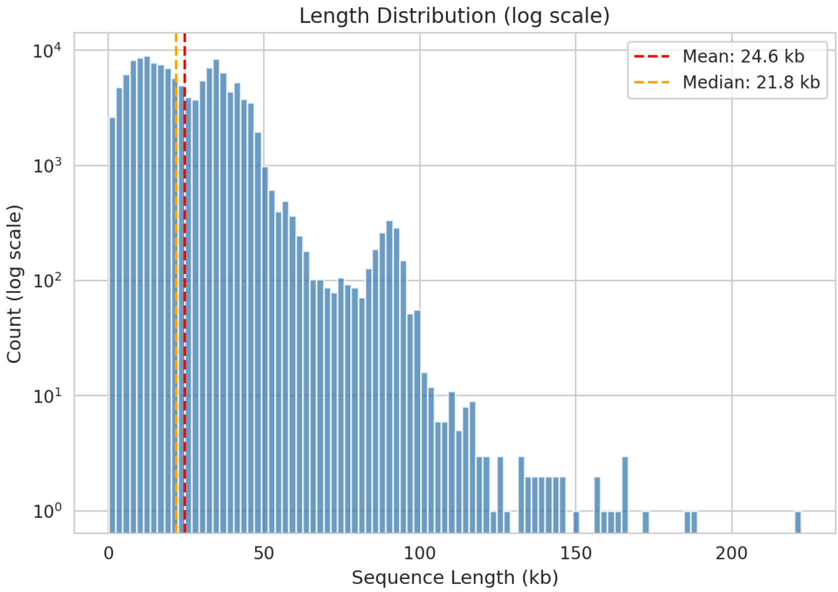
Full Prophage Homology Survey

MASH-based Analysis of 132,393 Prophage Sequences from 26,000 E. coli Genomes

132,393	8,187	24.6 kb	9,153,871
Total Sequences	Genomes	Mean Length	Similar Pairs (d<0.5)

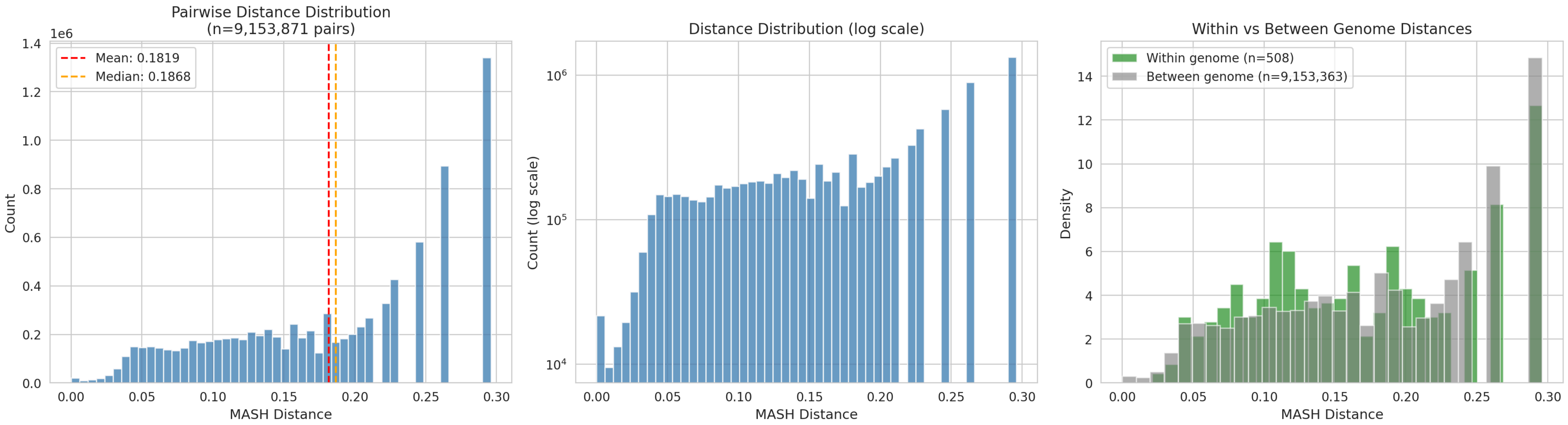
1. Sequence Length Distribution





Metric	Value
Count	132,393
Mean	24.6 kb
Median	21.8 kb
Std Dev	15.9 kb
Min	0.3 kb
Max	222.3 kb
Q25	11.9 kb
Q75	35.2 kb

2. MASH Distance Distribution



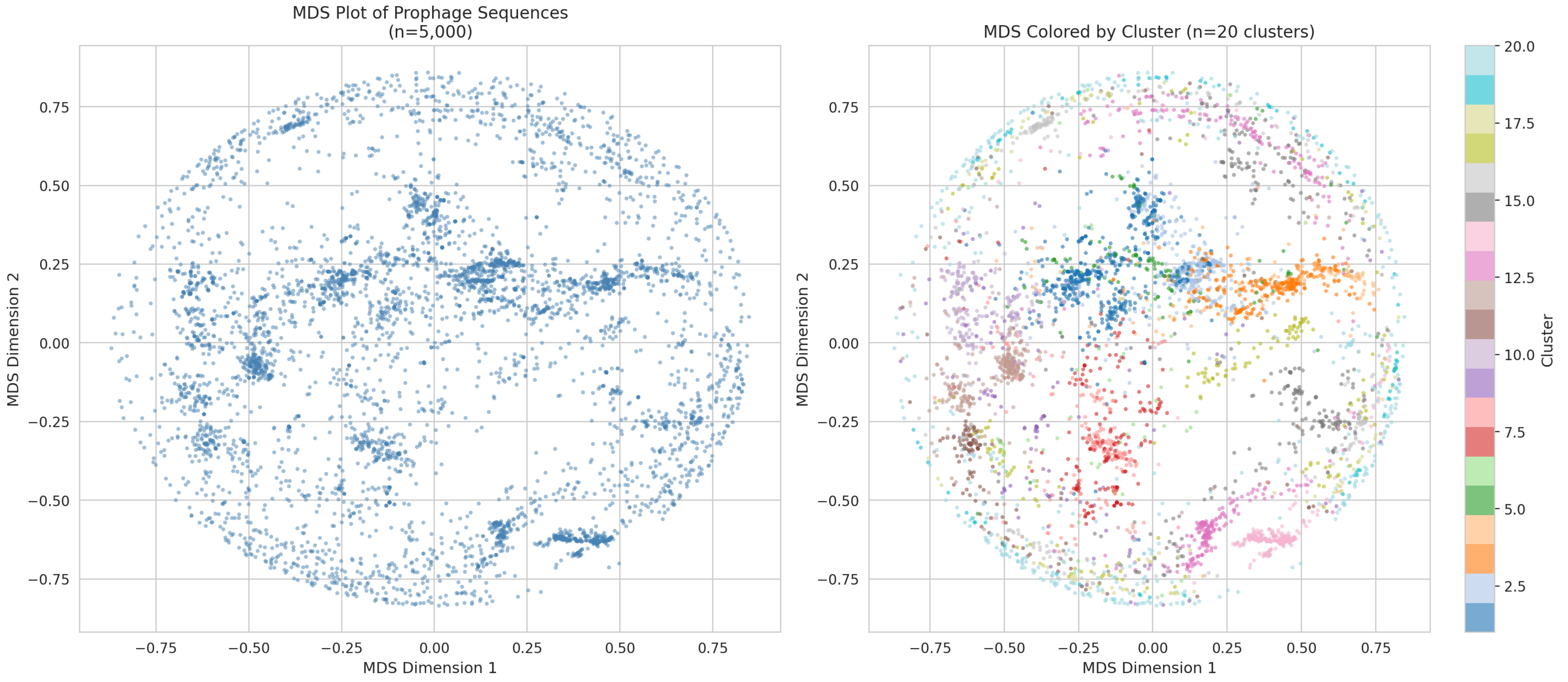
Metric	Overall
Pairs	9,153,871
Mean	0.1819
Median	0.1868
Std	0.0797

Metric	Within Genome
Pairs	508
Mean	0.1695
Median	0.1647
Std	0.0761

Metric	Between Genome
Pairs	9,153,363
Mean	0.1819
Median	0.1868
Std	0.0797

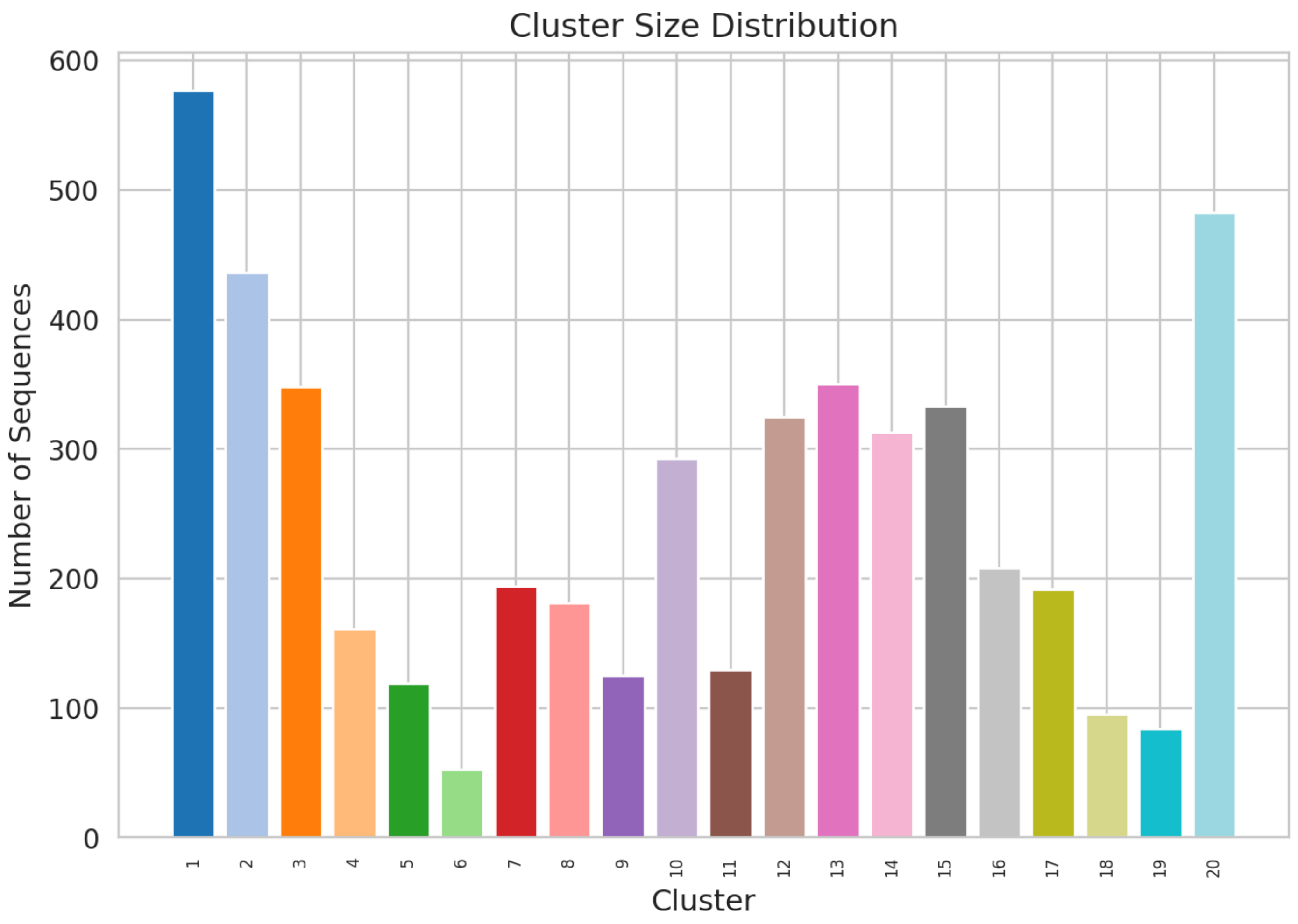
3. Multidimensional Scaling (MDS)

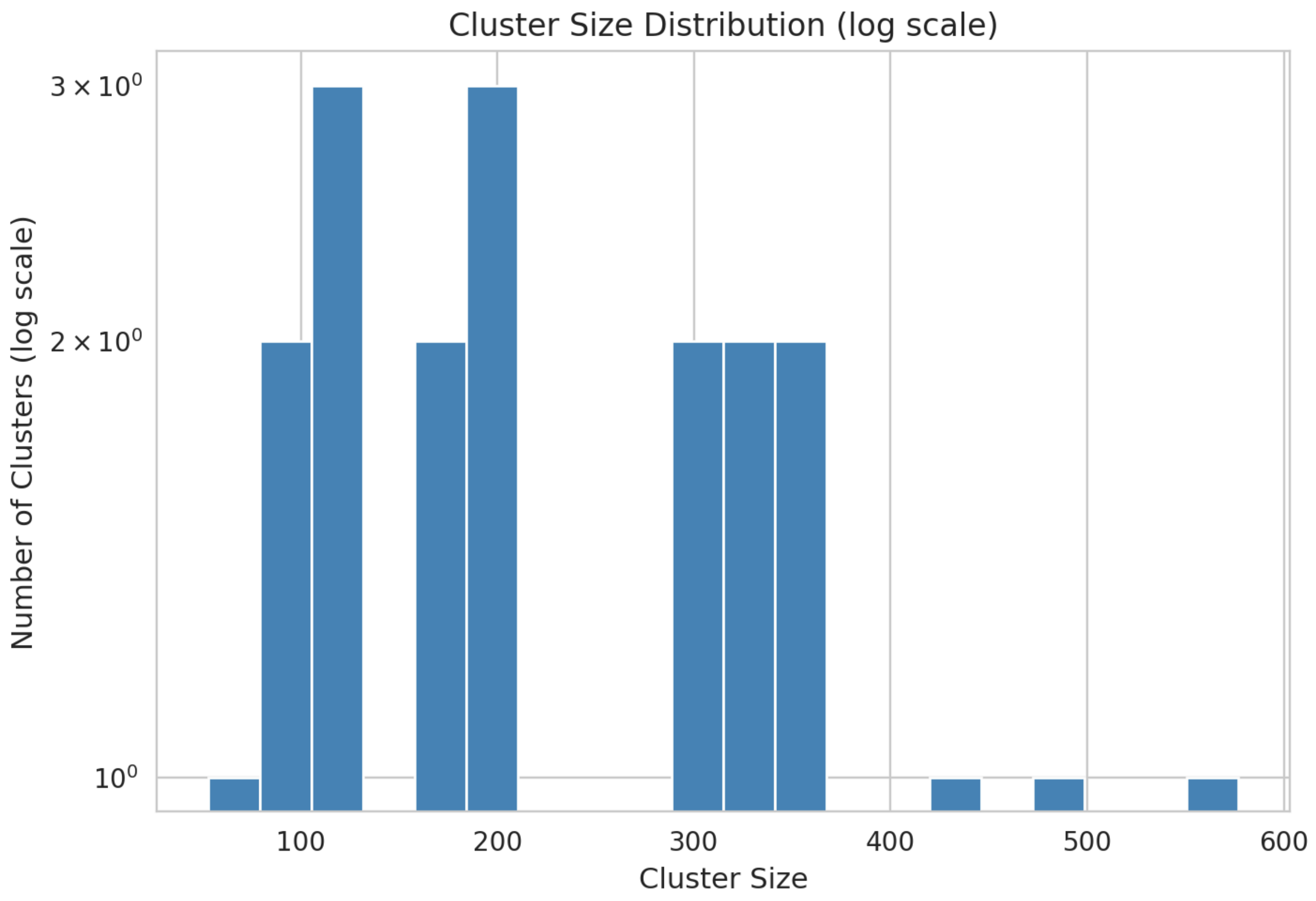
MDS stress: 1626006.7133

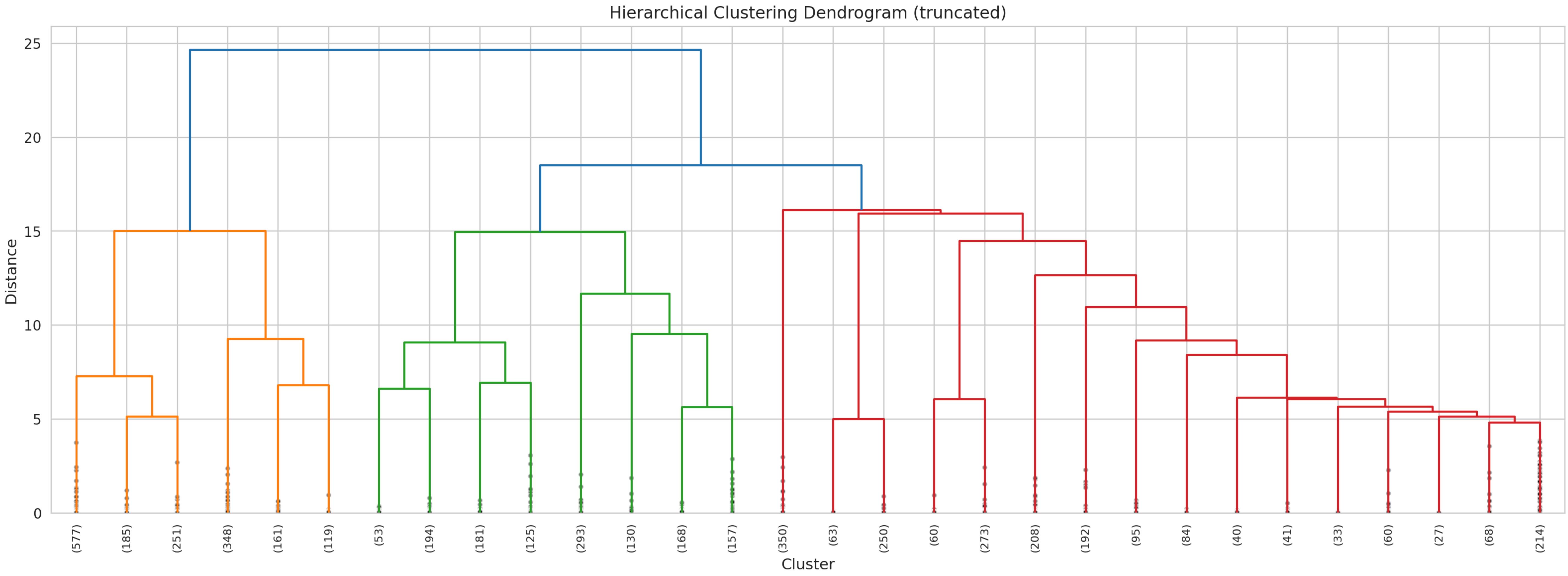


4. Hierarchical Clustering

Optimal number of clusters: 20

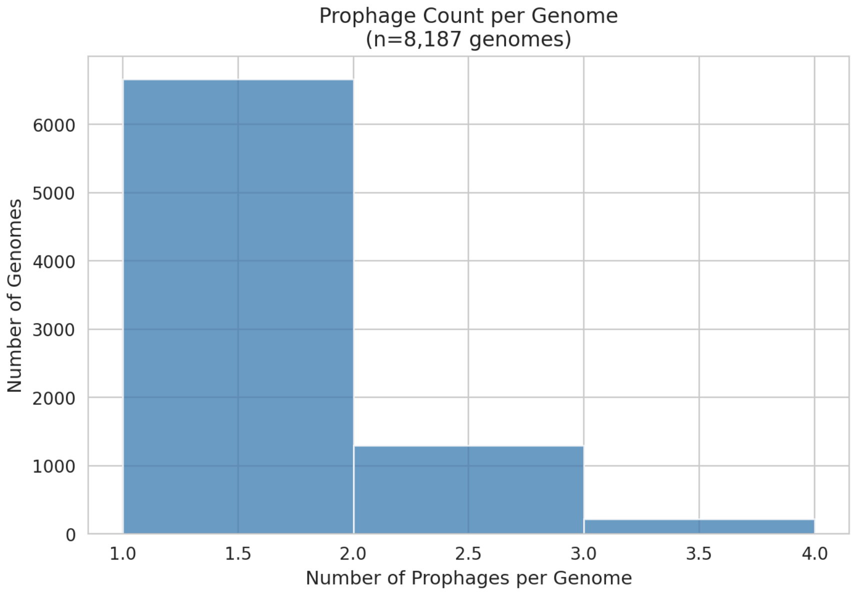


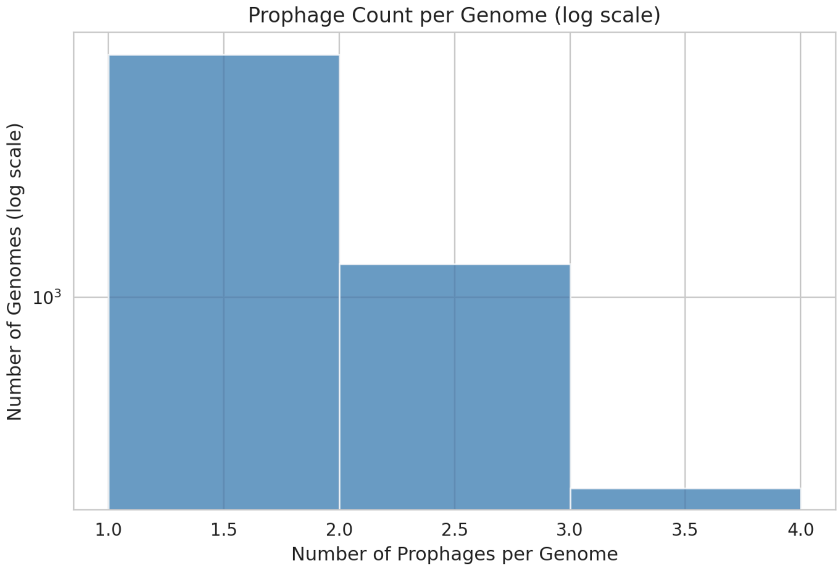




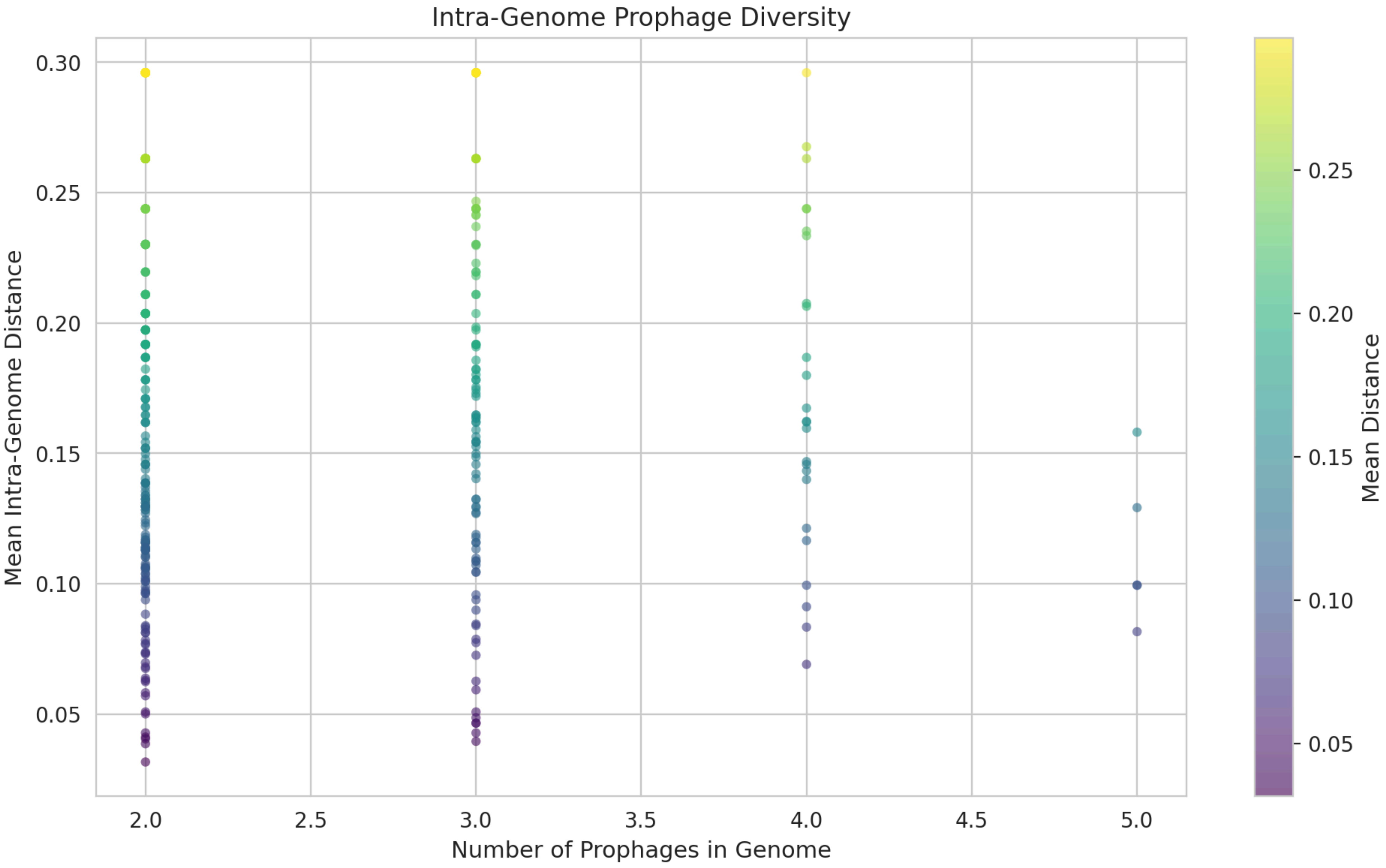
Hierarchical Clustering Dendrogram (truncated)
Distance
25
20
15
10
5
0
(577)
(185)
(251)
(348)
(161)
(119)
(53)
(194)
(181)
(125)
(293)
(130)
(168)
(157)
(350)
(63)
(250)
(60)
(273)
(208)
(192)
(95)
(84)
(40)
(41)
(33)
(60)
(27)
(68)
(214)
Cluster

5. Genome-Level Prophage Diversity





Metric	Value
Genomes with prophages	8,187
Genomes with 2+ prophages	1,527
Max prophages/genome	5
Mean prophages/genome	1.2



Intra-Genome Prophage Diversity
Mean Intra-Genome Distance
Number of Prophages in Genome
Mean Distance
0.30
0.25
0.20
0.15
0.10
0.05
2.0
2.5
3.0
3.5
4.0
4.5
5.0

6. Summary Statistics

Metric	Value	Notes
Total sequences	132,393	Full prophage set
Total genomes	8,187	From 26k E. coli genomes
Mean sequence length	24.6 kb	Range: 0.3-222.3 kb
Similar pairs ($d<0.5$)	9,153,871	Out of 9e+09 total pairs
Mean distance (similar pairs)	0.1819	Only pairs with $d<0.5$
MDS stress	1626006.7133	Lower is better
Number of clusters	20	Hierarchical clustering
Genomes with 2+ prophages	1,527	Intra-genome diversity

Generated by MASH v2.3 + Python analysis pipeline.
Analysis date: 2026-07-29. Sample size: 10,000 sequences for pairwise analysis.